

Solution of FA4 : Moment and Deviation

Problem 1 (Moments and (co)variances)

[Run (ii)] 1

设随机变量 X 为长度为 n 的随机 01 串（每一位以 p 的概率为 0，以 q 的概率为 1） S 的 runs 数量， $I_1, I_2, \dots, I_{n-1}$ 为指示随机变量， I_t 指示事件 $S_t \neq S_{t+1}$ 的发生，所以有 $X = 1 + \sum_{t=1}^{n-1} I_t$ 。对于任意 t ,

$$\text{Var}(I_t) = 2pq - (2pq)^2$$

$$\text{Cov}(I_t, I_{t+1}) = pq - (2pq)^2$$

由于不相邻的 I 之间独立，所以有

$$\text{Var}(X) = \sum_{t=1}^{n-1} \text{Var}(I_t) + \sum_{t=1}^{n-2} 2\text{Cov}(I_t, I_{t+1}) = (4n - 6)pq - (3n - 5)(2pq)^2$$

[Run (ii)] 2

设随机变量 X 为 h 个 0， t 个 1 组成的长度为 $n = h + t$ 的随机排列 π 中 0 的连续段的数量， $I_1, I_2, \dots, I_n$ 为指示随机变量， I_t 指示事件 $\pi_t = \pi_{t+1} = 0$ 的发生，所以有 $X = h - \sum_{t=1}^{n-1} I_t$ 。对于任意 t ,

$$\mathbb{E}(I_t) = \frac{h(h-1)}{n(n-1)} \triangleq p$$

$$\text{Var}(I_t) = \mathbb{E}(I_t^2) - \mathbb{E}(I_t)^2 = p - p^2$$

$$\text{Cov}(I_t, I_{t+1}) = \mathbb{E}(I_t I_{t+1}) - \mathbb{E}(I_t)\mathbb{E}(I_{t+1}) = \frac{h(h-1)(h-2)}{n(n-1)(n-2)} - p^2$$

所以有

$$\mathbb{E}(X) = h - \sum_{t=1}^{n-1} \mathbb{E}(I_t) = \frac{h(t+1)}{n}$$

$$\begin{aligned} \text{Var}(X) &= \sum_{t=1}^{n-1} \text{Var}(I_t) + \sum_{t=1}^{n-2} 2\text{Cov}(I_t, I_{t+1}) = \\ &= \frac{h(h-1)(n+2h-5)}{n(n-1)} - (3n-5) \left(\frac{h(h-1)}{n(n-1)} \right)^2 \end{aligned}$$

[Random sum (ii)]

根据全期望公式，可以知道

$$\mathbb{E}(S) = \mathbb{E}_N(\mathbb{E}(S|N)) = \mathbb{E}_N(\mu N) = \mu \mathbb{E}(N)$$

$$\mathbb{E}(S^2) = \mathbb{E}_N(\mathbb{E}(S^2|N)) = \mathbb{E}_N(\mu^2 N^2 + \sigma^2 N) = \mu^2 \mathbb{E}(N^2) + \sigma^2 \mathbb{E}(N)$$

带入可得

$$\text{Var}(S) = \mathbb{E}(S^2) - \mathbb{E}^2(S) = \mu^2 \text{Var}(N) + \sigma^2 \mathbb{E}(N)$$

[Maximum variance]

设 $\mathbb{E}(Y) = p_1 + p_2 + \cdots + p_n = N$, 根据柯西不等式有

$$\text{Var}(Y) = N - \sum_{k=1}^n p_k^2 \leq N - \left(\sum_{k=1}^n 1 \cdot p_k \right)^2 / \left(\sum_{k=1}^n 1^2 \right) = N - N^2/n$$

而当所有 $p_k = N/n$ 时等号恰好能取到。

[Moments (i)]

当 $\alpha \geq 1$ 时, $\mathbb{E}(X^\alpha) \geq \sum_{n=1}^{\infty} \frac{1}{n+1} \rightarrow \infty$ 发散。

当 $\alpha < 1$ 时, $\mathbb{E}(X^\alpha) \leq \sum_{n=1}^{\infty} \frac{1}{n^{2-\alpha}} \leq 1 + \int_1^{\infty} x^{\alpha-2} dx \leq 1 + \frac{1}{1-\alpha}$ 收敛。

[Moments (ii)]

$X \sim \text{Geo}(p)$

$$\mathbb{E}(X^3) = p \sum_{n=1}^{\infty} n^3 (1-p)^{n-1} = \frac{p^2 - 6p + 6}{p^3}$$

$$\mathbb{E}(X^4) = p \sum_{n=1}^{\infty} n^4 (1-p)^{n-1} = \frac{-p^3 + 14p^2 - 36p + 24}{p^4}$$

[Covariance and correlation (i)]

从 $X = aY + b$ 证明 $\rho^2 = 1$ 的方向是容易的, 考虑证明另外一个方向。设 μ 为 (X, Y) 的联合分布, $x \triangleq \mathbb{E}(x)$, $y \triangleq \mathbb{E}(y)$

$$\rho^2 = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)\text{Var}(Y)} = \frac{\left(\mathbb{E}_{(X,Y) \sim \mu} (X-x)(Y-y) \right)^2}{\mathbb{E}_{X \sim \mu} (X-x)^2 \mathbb{E}_{(Y) \sim \mu} (Y-y)^2}$$

根据柯西不等式, 上式小于等于 1, 且等号仅在对于任意 $(X, Y) \sim \mu$ 时, $\frac{X-x}{Y-y}$ 恒等于某个常数 k 时取得, 此时 $X = kY - ky + x$ 。

[Covariance and correlation (ii)]

设 $U = X + Y, V = Y + Z, W = Z - X$, X, Y, Z 为互相独立的非常数随机变量

$$\text{Cov}(U, V) = \text{Var}(Y) > 0$$

$$\text{Cov}(V, W) = \text{Var}(Z) > 0$$

$$\text{Cov}(U, W) = -\text{Var}(X) < 0$$

[Berkson's fallacy]

对于 $g \in G$, 设 X_g 为指示随机变量表示 g 住院, Y_g 为指示随机变量表示 g 住院并且感染了疾病 C。可以知道 $X = \sum_{g \in G} X_g, Y = \sum_{g \in G} Y_g$, 由于 (X_g, Y_g) 是独立同分布, 以及

- $\mathbb{E}(X_g) = \mathbb{E}(X_g^2) = a + \gamma c - a\gamma c, \mathbb{E}(Y_g) = \mathbb{E}(Y_g^2) = \gamma p;$
- $\mathbb{E}(X_g Y_g) = \mathbb{E}(Y_g)$

所以

$$\begin{aligned}\rho(X, Y) &= \rho(X_g, Y_g) = \frac{\mathbb{E}(Y_g) - \mathbb{E}(X_g)\mathbb{E}(Y_g)}{\sqrt{(\mathbb{E}(X_g) - \mathbb{E}(X_g)^2)(\mathbb{E}(Y_g) - \mathbb{E}(Y_g)^2)}} \\ &= \sqrt{\frac{\mathbb{E}(Y_g)(1 - \mathbb{E}(X_g))}{\mathbb{E}(X_g)(1 - \mathbb{E}(Y_g))}} \\ &= \sqrt{\frac{\gamma p(1 - a)(1 - \gamma c)}{(1 - \gamma p)(a - \gamma c - a\gamma c)}}\end{aligned}$$

[Correlation and independence]

设 $U = X + Y, V = |X - Y|$, 其中 $X, Y \sim \text{Bernoulli}(1/2)$ 。

依赖性: 当 $\Pr(U = 1|V = 1) = 1 > \Pr(U = 1)$ 。

不相关性: $\mathbb{E}(U) = 1, \mathbb{E}(V) = 1/2, \mathbb{E}(UV) = 1/2$, 所以 $\text{Cov}(U, V) = 0$ 。

[Conditional variance formula]

定义

$$\begin{aligned}\text{Var}_\mu(Y|X) &\triangleq \mathbb{E}_{Y \sim \mu(\cdot|X)} \left(Y - \mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2 \\ &= \mathbb{E}_{Y \sim \mu(\cdot|X)} Y^2 - \left(\mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2\end{aligned}$$

根据全期望公式

$$\mathbb{E}_{X \sim \mu} \mathbb{E}_{Y \sim \mu(\cdot|X)} f(Y) = \mathbb{E}_\mu f(Y)$$

有

$$\begin{aligned}\text{Var}_\mu(Y) &= \mathbb{E}_\mu Y^2 - (\mathbb{E}_\mu Y)^2 \\ &= \mathbb{E}_{X \sim \mu} \mathbb{E}_{Y \sim \mu(\cdot|X)} Y^2 - \left(\mathbb{E}_{X \sim \mu} \mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2 \\ &= \mathbb{E}_{X \sim \mu} \mathbb{E}_{Y \sim \mu(\cdot|X)} Y^2 - \mathbb{E}_{X \sim \mu} \left(\mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2 \\ &\quad + \mathbb{E}_{X \sim \mu} \left(\mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2 - \left(\mathbb{E}_{X \sim \mu} \mathbb{E}_{Y \sim \mu(\cdot|X)} Y \right)^2 \\ &= \mathbb{E}_{X \sim \mu} \text{Var}_\mu(Y|X) + \text{Var}_{X \sim \mu} \mathbb{E}_\mu(Y|X)\end{aligned}$$

[Conditional covariance]

定义

$$\begin{aligned}\text{Cov}_\mu(X, Y|Z) &\triangleq \mathbb{E}_{(X,Y) \sim \mu(\cdot|Z)} \left(X - \mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(Y - \mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right) \\ &= \mathbb{E}_{(X,Y) \sim \mu(\cdot|Z)} Y^2 - \left(\mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(\mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right)\end{aligned}$$

同理根据全期望公式

$$\mathbb{E}_{Z \sim \mu} \mathbb{E}_{(X,Y) \sim \mu(\cdot|Z)} f(X, Y) = \mathbb{E}_\mu f(X, Y)$$

有

$$\begin{aligned}\text{Cov}_\mu(X, Y) &= \mathbb{E}_\mu XY - (\mathbb{E}_\mu X)(\mathbb{E}_\mu Y) \\ &= \mathbb{E}_{Z \sim \mu} \mathbb{E}_{(X,Y) \sim \mu(\cdot|Z)} XY - \left(\mathbb{E}_{Z \sim \mu} \mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(\mathbb{E}_{Z \sim \mu} \mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right) \\ &= \mathbb{E}_{Z \sim \mu} \mathbb{E}_{(X,Y) \sim \mu(\cdot|Z)} XY - \mathbb{E}_{Z \sim \mu} \left(\mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(\mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right) \\ &\quad + \mathbb{E}_{Z \sim \mu} \left(\mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(\mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right) - \left(\mathbb{E}_{Z \sim \mu} \mathbb{E}_{(X,\cdot) \sim \mu(\cdot|Z)} X \right) \left(\mathbb{E}_{Z \sim \mu} \mathbb{E}_{(\cdot,Y) \sim \mu(\cdot|Z)} Y \right) \\ &= \mathbb{E}_{Z \sim \mu} \text{Cov}_\mu(X, Y|Z) + \text{Cov}_{Z \sim \mu}(\mathbb{E}_\mu(X|Z), \mathbb{E}_\mu(Y|Z))\end{aligned}$$

Problem 2 (Markov and Chebyshev)

[Markov's inequality]

注意到 $X \geq x \Leftrightarrow e^{\beta X} \geq e^{\beta x}$ 。

由于 $e^{\beta X}$ 是一个非负的随机变量, $e^{\beta x} > 0$, 所以可以对 $Y = e^{\beta X}$ 应用 Markov's inequality:

$$\Pr[X \geq x] = \Pr[e^{\beta X} \geq e^{\beta x}] \leq \frac{\mathbb{E}[e^{\beta X}]}{e^{\beta x}}$$

[Chebyshev's inequality (i) (Cantelli's inequality)]

令 $Y = X - \mathbb{E}[X]$, 有 $\mathbb{E}[Y] = 0$ 以及 $\text{Var}[Y] = \text{Var}[X]$

对于任意 $t, u > 0$, 考虑 Markov's inequality

$$\Pr[Y \geq t] = \Pr[(Y + u)^2 \geq (t + u)^2] \leq \frac{\mathbb{E}[(Y + u)^2]}{(t + u)^2} = \frac{\text{Var}[X] + u^2}{(t + u)^2} := \varphi(u)$$

令 $\varphi'(u) = 0$, 得到 $\min \varphi(u) = \varphi(\text{Var}[X]/t) = \frac{\text{Var}[X]}{\text{Var}[X] + t^2}$ 。故 $\Pr[Y \geq t] \geq \frac{\text{Var}[X]}{\text{Var}[X] + t^2}$ 。

[Chebyshev's inequality (ii) (Paley–Zygmund inequality)]

对于任意 $0 \leq a \leq 1$, 我们可以将随机变量 X 写作 $X = X \cdot (\mathbf{1}_{X \leq a \cdot \mathbb{E}[X]} + \mathbf{1}_{X > a \cdot \mathbb{E}[X]})$ 。取期望可以得到

$$\mathbb{E}[X] = \mathbb{E}[X \cdot \mathbf{1}_{X \leq a \cdot \mathbb{E}[X]}] + \mathbb{E}[X \cdot \mathbf{1}_{X > a \cdot \mathbb{E}[X]}]$$

一方面, $\mathbb{E}[X \cdot \mathbf{1}_{X \leq a \cdot \mathbb{E}[X]}] \leq a \cdot \mathbb{E}[X]$

另一方面, 由 [Cauchy–Schwarz inequality](#),

$$\mathbf{E}[X \cdot \mathbf{1}_{X > a \cdot \mathbf{E}[X]}] \leq \sqrt{\mathbf{E}[X^2] \mathbf{E}[\mathbf{1}_{X > a \cdot \mathbf{E}[X]}]} = \sqrt{\mathbf{E}[X^2] \Pr[X > a \cdot \mathbf{E}[X]]}$$

代入整理可以得到

$$\Pr[X > a \cdot \mathbf{E}[X]] \geq \frac{(1-a)^2 \mathbf{E}[X]^2}{\mathbf{E}[X^2]}$$

上式中令 $a = 0$, 可以得到

$$\Pr[X = 0] = 1 - \Pr[X > 0] \leq 1 - \frac{\mathbf{E}[X]^2}{\mathbf{E}[X^2]} = \frac{\mathbf{Var}[X]}{\mathbf{E}[X^2]} \leq \frac{\mathbf{Var}[X]}{\mathbf{E}[X]^2}$$

将上式应用到一系列 indicator 之和时, 记 $X = \sum_{r=1}^n X_r = \sum_{r=1}^n I_{A_r}$, $\sum^*$ 枚举了所有彼此不独立的有序对 A_r, A_s (也即交换 r, s 我们认为是不同的 pair)。

计算 $\mathbf{Var}[X]$:

$$\begin{aligned} \mathbf{Var}[X] &= \sum_{r=1}^n \mathbf{Var}[I_{A_r}] + \sum^* \mathbf{Cov}[I_{A_r}, I_{A_s}] \\ &= \sum_r (\mathbf{E}[I_{A_r}^2] - \mathbf{E}[I_{A_r}]^2) + \sum^* (\mathbf{E}[I_{A_r} I_{A_s}] - \mathbf{E}[I_{A_r}] \mathbf{E}[I_{A_s}]) \\ &= \sum_r (\Pr[A_r] - \Pr[A_r]^2) + \sum^* (\Pr[A_r \cap A_s] - \Pr[A_r] \Pr[A_s]) \\ &\leq \sum_r \Pr[A_r] + \sum^* \Pr[A_r \cap A_s] \\ &= \mathbf{E}[X] + \sum^* \Pr[A_r \cap A_s] \end{aligned}$$

所以

$$\Pr[X = 0] \leq \frac{\mathbf{Var}[X]}{\mathbf{E}[X]^2} \leq \frac{1}{\mathbf{E}[X]} + \frac{1}{\mathbf{E}[X]^2} \sum^* \Pr[A_r \cap A_s]$$

Problem 3 (Random graphs)

[Common neighbor]

对于两个点 u, v , 他们有公共邻居的事件记为 $C_{u,v}$, 那么 $\Pr[\overline{C_{u,v}}] = (1 - p^2)^{n-2}$

进而 $\Pr[\exists u, v \in V, \overline{C_{u,v}}] \leq \binom{n}{2} \Pr[\overline{C_{u,v}}] = \binom{n}{2} (1 - p^2)^{n-2} \rightarrow 0$ (when $n \rightarrow \infty$), 故

$$\Pr[\forall u, v \in V, C_{u,v}] = 1 - \Pr[\exists u, v \in V, \overline{C_{u,v}}] \rightarrow 1$$

也可以记没有公共邻居的点对数为 C , 有 $\mathbf{E}[C] = \sum_{u,v} \Pr[\overline{C_{u,v}}] = \binom{n}{2} (1 - p^2)^{n-2}$, 由 Markov's inequality 可以得到

$$\Pr[C \geq 1] \leq \frac{\mathbf{E}[C]}{1} = \binom{n}{2} (1 - p^2)^{n-2} \rightarrow 0 \text{ (when } n \rightarrow \infty)$$

也即 $\Pr[C = 0] \rightarrow 1$

[Isolated vertices]

记 $X_v = \mathbf{1}[v \text{ 是一个孤立点}]$, $X = \sum_v X_v$ 为 $G(n, p)$ 中孤立点的期望个数, 那么有 $\mathbf{E}[X] = n(1-p)^{n-1}$

令 $p = c \ln n / n$, 那么有 $\lim_{n \rightarrow \infty} \mathbf{E}[X] = n(1 - \frac{c \ln n}{n})^n = \lim_{n \rightarrow \infty} n e^{-c \ln n} = \lim_{n \rightarrow \infty} n^{1-c}$

一方面, 若 $c > 1$, 由Markov's inequality, $\Pr[X \geq 1] \leq \frac{\mathbf{E}[X]}{1} \rightarrow 0$ 。也即 $p = \omega(\log n / n)$ 时, $G(n, p)$ 不含孤立点的概率几乎为 1。

另一方面, 当 $c < 1$ 时, 孤立点的期望个数趋于无穷, 但这并不能说明"几乎所有的图都含有孤立点" (可能有少部分的图含有大量孤立点, 而其余不含孤立点, 我们需要排除这种情况)。一种经典的方法是考虑 X 的二阶矩 (second moment argument), 我们有

$$\mathbf{E}[X^2] = \mathbf{E}\left[\left(\sum_v X_v\right)^2\right] = \sum_v \mathbf{E}[X_v^2] + \sum_{u \neq v} \mathbf{E}[X_u X_v]$$

对于 $\mathbf{E}[X_v^2]$, 因为 $X_v \in \{0, 1\}$, 所以 $X_v^2 = X_v$, 进而 $\mathbf{E}[X_v^2] = \mathbf{E}[X_v]$ 。

对于 $\mathbf{E}[X_u X_v]$, u, v 相邻的边共有 $2(n-1) - 1 = 2n - 3$ 条, 所以 $\mathbf{E}[X_u X_v] = \Pr[A_u \cap A_v] = (1-p)^{2n-3}$ 。

由 Chebyshev's inequality (或者直接应用上一题中的Paley-Zygmund inequality)

$$\Pr[X = 0] \leq \Pr[|X - \mathbf{E}[X]| \geq \mathbf{E}[X]] \leq \frac{\mathbf{E}[X^2] - \mathbf{E}[X]^2}{\mathbf{E}[X]^2}$$

进而 $n \rightarrow \infty$ 时

$$\Pr[X = 0] \leq \frac{\mathbf{E}[X^2]}{\mathbf{E}[X]^2} - 1 \leq \frac{1}{n^{1-c}} + \frac{n(n-1)(1-p)^{2n-3}}{n^2(1-p)^{2n-2}} - 1 = \frac{1}{n^{1-c}} + (1 - \frac{1}{n}) \frac{1}{1 - c \frac{\ln n}{n}} - 1 \rightarrow 0$$

所以 $c < 1$ 时 (或者说 $p = o(\log n / n)$ 时), 以接近 0 的概率, 图中不存在孤立点。